

Allison M. Barry, PhD

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Nationality: Canadian

Current position

*2023 - **Postdoctoral Research Associate**
University of Texas at Dallas, Department of Neuroscience (Prof Theodore Price)
Center for Advanced Pain Studies
NIH Precision Human Pain Network
University of Texas at Dallas; USA

**Employed as remote contractor with UTDallas Sept 2023- Jan 2026*

Education

2017-2022 **DPhil. (PhD), Ion Channels in Health and Disease**
University of Oxford, Nuffield Department of Clinical Neurosciences
Oxford, UK

- Thesis: Exploring the molecular underpinnings of neuropathic pain in primary afferents
- Advisors: Prof. David Bennett, Dr. Georgios Baskozos, Prof Stephen Tucker

2015-2017 **MSc., Neurosciences**
IMPRS, Max-Planck-Institut Für Experimentelle Medizin
Göttingen, DE

- Thesis: Modulation of TRPV1 with interaction partners
- Advisor: Prof. Manuela Schmidt

2011-2015 **BSc., Neuroscience, First Class Honours**
Dalhousie University
Halifax, CA

- Thesis Advisor: Prof. Victor Rafuse, Department of Medical Neurosciences

Prior professional experience

2023 - **Academic Visitor**
University of Oxford, NDCN (Prof David Bennett)
Oxford, UK

2023-2025 **Postdoctoral Neuroscientist**
Department of Pharmacology (Prof Manuela Schmidt)
University of Vienna; AT

2022-2023 **Postdoctoral Neuroscientist**
University of Oxford, NDCN (Prof David Bennett)
PAINSTORM, Partnership for Assessment and Investigation of Neuropathic Pain: Studies
Tracking Outcomes, Risks and Mechanisms
Oxford, UK

- 2017-2018 **wiss. Mitarbeiter*innen**
Max-Planck-Institut Für Experimentelle Medizin
Göttingen, DE
- 2012-2015 **Research Assistant**
Dalhousie University, Dept Psychology and Neuroscience
Halifax, CA

Grant History

- 2017-2022 Lead Applicant. Wellcome Trust. Four-Year PhD Studentship in Basic Science + supplementary Costs. University of Oxford, UK. Grant ID: 215145/Z/18/Z. **188014 GBP**
- 2015-2016 Project Lead. NSERC. Characterizing Synaptopodin-2 during skeletal myogenesis using embryonic stem cell derived muscle. Dalhousie University, CA. Grant ID: 581929. **4500 CAD**
- 2014-2015 Project Lead. NSERC. Analysis of visual system neurons in Drosophila. Dalhousie University, CA. Grant ID: 560615. **4500 CAD**
- 2013-2014 Project Lead. NSERC. Neurons of the fly's visual system. Dalhousie University, CA. Grant ID: 540051 . **4500 CAD**

Selected Awards

Honours

- 2025 Original Research selected as Editor's Choice (PAIN)
- 2023 Original Research selected as Editor's Choice (PAIN)
- 2022 Abstract selected for presentation, NeuPSIG Symposium, IASP World Congress
- 2021 Abstract selected for presentation, IASP Euro Pain School

Scholarships/Fellowships

- 2017-2022 Medical Sciences DTC Doctoral Scholarship, Green Templeton. **60560 GBP**
- 2015-2017 MSc Stipend; International Max Planck Research School, Göttingen. **14400 EUR**
- 2015 Dalhousie in-course Scholarship, Dalhousie University
- 2014 Dr. M.G. Hickman Memorial Scholarship, Dalhousie University
- 2013 Marjorie F Ellis Scholarship, Dalhousie University
- 2012 Dalhousie Entrance Scholarship, Dalhousie University

Student Supervision

- BSc Joshua Diaz (UTDallas, 2026-)
Na Zhao (University of Oxford, 2021-2024)
Anna Klishina (University of Oxford, 2019)

- MSc Maximilian Mayrhauser (University of Vienna, 2023-2024)
Thesis: Benchmarking data-independent acquisition mass spectrometry analyses for metaproteomic data (co-supervised with Manuela Schmidt)
- PhD Marisol Mancilla Moreno (University of Texas at Dallas, 2022 - present)
Thesis: Genomic and transcriptomic divergence in dorsal root ganglia (co-supervised with Ted Price)

Teaching

- 2024-2025 **Seminar Series**
Bioinformatics for Biologists (postgraduate + postdoctoral audience).
Dept of Pharmacology, University of Vienna; AT
Organized, planned, and administered a seminar series for biologists looking to analyze omics data. Topics centered around version controlling, R basics, resource development, and debugging
- 2022 **Seminar Speaker**
Introduction to „omics“ (postgraduate seminar series).
Dept Clinical Neurosciences, University of Oxford, UK
Responsibility include helping design seminar series, developing and giving lecture material
- 2014-2015 **Laboratory Teaching Assistant**
Genetics and Molecular Biology (5x 20 undergraduate students).
Dalhousie University; Dept Biology. CA.
Responsibilities included building + presenting introductory lecture materials, leading discussions, supervising laboratory work for 2 year undergraduate students, grading assignments, and meeting with students individually
- 2014-2016 **Tutor (private)**
Genetics and Molecular Biology, Molecular Neuroscience
Student mentorship and one-on-one teaching across undergraduate neuroscience and biology courses for students at Dalhousie University in BSc Neuroscience, Biology, and Biochemistry programs

Society Membership

- 2025- European Bioinformatics Community for MS (EuBIC-MS)
2024- Peripheral Nerve Society (PNS)
2019- International Association for the Study of Pain (IASP)

Scientific Community Contributions

- ongoing Peer Reviewer, eg. PAIN; Nat Neuro, JPNS, Eur J Pain, Bioinformatics Advances
- ongoing Mentor, Disabled in STEM (<https://disabledinstem.wordpress.com/>)

ongoing	Database management for FAIR omics data, https://livedataoxford.shinyapps.io/drg-directory/
ongoing	Outreach, registered “Skype a Scientist”
2018-2020	Outreach, Oxford Hands on Science committee member (http://oxhos.org/)
2018	Outreach, Oxford Brain Challenge Organizer, St. Hilda’s College (Oxford, UK)

Presentations

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| 2026 | Dissecting pathology using human DRG spatial proteomics. PNS Annual Meeting, (Maastricht, NL) |
| | Multi-omic hDRG dysregulation in Opioid Use Disorder. Pain Mechanisms Conference, (Verona, IT) |
| 2025 | Applying ML algorithms for the study of neuropathic pain (Workshop). NeuPSIG, (Berlin, DE) |
| | AI in Exploratory Human Research. IASP Seminar Series, Sept (online) |
| | Multi-omics in pain as a foundation for precision medicine. Dalhousie University. (Halifax, CA) |
| 2024 | Predicting “pain genes”: multi-modal data integration using probabilistic classifiers and interaction networks. Canadian Neuroscience Society (CNS) Seminar Series, 2024 (online) |
| | Machine Learning and AI in Basic and Clinical Pain Research: A Primer on Big Data Methods (Masterclass). IASP World Congress, 2024 (Amsterdam, NL) |
| | Deep RNA-seq of male and female murine sensory neuron subtypes after nerve injury. Pathway to Publication and Impact, IASP Early Career Researcher Webinar, 2024 (online) |
| 2023 | Predicting “pain genes”. Mathematical Biology Series, Open University, 2023 (London, UK) |
| | “Omics” databases for data integration in pain. Advanced Pain Discovery Platform Conference, 2023 (Nottingham, UK) |
| 2022 | Deep sequencing of murine DRG subpopulations after nerve injury. NeuPSIG Symposium, IASP World Congress, 2022 (Toronto, CA) |
| | Aggregating sequencing data in pain research. GW4 3Rs Symposium, 2022 (Bath, UK) |

Exploring the molecular underpinnings of neuropathic pain in primary afferent subtypes. Thomas Willis Day, 2022 (Oxford, UK)

- 2021 Exploring the molecular underpinnings of neuropathic pain in primary afferent subtypes. IASP Euro Pain School, 2021 (online)

Aggregating sequencing data in pain research. Oxford NC3Rs Symposium, 2021 (Oxford, UK)

Conference Abstracts

Only first/last author included

- 2025 **Allison Barry**, Julia Sondermann, Joseph Lesnak, Feng Xian, Úrzula Franco-Enzástiga, Jayden O'Brien, David Gomez Varela, Morgan Schackmuth, Stephanie Shiers, Theodore Price, Manuela Schmidt. Multi-omic integration with human DRG proteomics highlights TNF α signalling as a relevant sexually dimorphic pathway. APMRS (Vienna, AT)

Allison Barry, Stephanie Shiers, Chris Hsu, Michael MacCoss, Theodore Price. Human DRG spatial proteomics in diabetic peripheral neuropathy. NeuPSIG (Berlin, DE)

Allison Barry, Julia Sondermann, Feng Xian, Thomas Haberl, David Gomez-Varela, Fritz Benseler, Noa Lipstein, Nils Brose, Manuela Schmidt. Proximity labelling reveals the compartmental proteome of murine sensory neurons. NeuPSIG (Berlin, DE)

Allison Barry, Julia Sondermann, Joseph Lesnak, Feng Xian, Úrzula Franco-Enzástiga, Jayden O'Brien, David Gomez Varela, Morgan Schackmuth, Stephanie Shiers, Theodore Price, Manuela Schmidt. Multi-omic integration with human DRG proteomics highlights TNF α signalling as a relevant sexually dimorphic pathway. Peripheral Nerve Society, 2025 (Edinburgh, UK)

- 2024 **Allison Barry**, Julia Sondermann, Feng Xian, Morgan Schackmuth, Stephanie Shiers, Theodore Price, Manuela Schmidt. Toward Defining the Human DRG Proteome in Females and Males. IASP World Congress, 2024 (Amsterdam, NL)

- 2023 **Allison Barry**, Na Zhao, David Bennett, Georgios Baskozos. Predicting “pain genes”: multi-modal data integration using probabilistic classifiers and interaction networks. Advanced Pain Discovery Platform Congress, 2023 (Nottingham, UK)

Na Zhao, David Bennett, Georgios Baskozos, **Allison Barry**. Pain RNAseq Hub: A Shiny-Based Database for Interactive Visualisation and Network Analyses of RNA-seq data in Neuropathic Pain. NeuPSIG, 2023 (Lisbon, PG)

- 2022 **Allison Barry**, Na Zhao, David Bennett, Georgios Baskozos. Deep Sequencing of Murine DRG Subpopulations after Nerve Injury. IASP World Congress, 2022 (Toronto, CA)
- 2019 **Allison Barry**, Steven Middleton, David Bennett. Exploring the nociceptor transcriptome after nerve injury. Thomas Willis Day, 2019 (Oxford, UK)
- Allison Barry**, Steven Middleton, David Bennett. Exploring primary afferent transcriptome changes in neuropathic pain. OXION: Ion Channels and Disease Initiative Day, 2019 (Oxford, UK)

Selected preprints

Sankaranarayanan I, Mazhar K, **Barry AM**, Shiers S, Wangzhou A, Sondermann JR, Xian F, Inturi N, Wilde MA, Meyers EC, Lesnak JB, O'Brien JA, Yousuf MS, Schmidt M, Dussor G, Vines E, Horton P, Cervantes A, Khan T, Funk G, Tavares-Ferreira D, PRECISION Human Pain Network, Dougherty PM, Curatolo M, Price TJ (2026). Progressive neurodegeneration in human dorsal root ganglion from diabetes to painful neuropathy. *in review*. <https://doi.org/10.64898/2026.01.16.700028>

Journal Publications

supervised students underlined

Sondermann JS*, **Barry AM***, Xian F, Haberl T, Gomez-Varela D, Benseler F, Brose N, Lipstein N, Schmidt M (2026). Proximity labelling reveals the compartmental proteome of murine sensory neurons. *European Journal of Pain*. <https://doi.org/10.1101/2025.06.13.659475> *shared first authorship

Wang S, **Barry AM**, Young N, Lee YK, Shim SW, Wang X, Kim HC, Gonzalez-Cano R, Stirling-Barros L, Costigan M, Baskozos G, Bennett DLH, Rinaldi S, Bae Oh S, Davies AJ (2025). NKG2D receptor ligands are cell surface biomarkers for injured murine and human nociceptive neurons. *Journal of Neuroinflammation*. <https://doi.org/10.1186/s12974-025-03675-1>

Grundtner S, Xian F, Malzl D, Radits T, **Barry AM**, Sondermann JR, Menche J, Gómez-Varela D, Schmidt M (2025). Temporal and demographic patterns of peripheral nerve proteomes in preclinical neuropathic pain. *Biomedicine & Pharmacotherapy*. DOI: 10.1016/j.biopha.2025.118855

Arendt-Tranholm A, Sankaranarayanan I, Payne C, Mancilla Moreno M, Mazhar K, Yap N, Chiu AP, **Barry AM**, Patel PP, Inturi NN, Tavares Ferreira T, Amin A, Karandikar M, Jarvik JG, Turner JA, Hofstetter CP, Curatolo M, Price TJ (2025). Single-cell characterization of the human C2 dorsal root ganglion recovered from C1-2 arthrodesis surgery: implications for neck pain. *Brain*, awaf370, <https://doi.org/10.1093/brain/awaf370>

Copits BA, Curatolo M, Dougherty PM, Gereau IV RW, Luo W, Martone M, Olausson H, Price TJ, Renthal W, Woolf CJ, Zhao G, **NIH PRECISION Human Pain Network** (2025). Human pain neuroscience and the next generation of pain therapeutics. *Neuron*. 113(9). p1304-1306. <https://doi.org/10.1016/j.neuron.2025.04.005>. **Featured Article**.

Barry AM, Sondermann JS, Lesnak JB, Xian F, Franco-Enzastiga U, O'Brien JA, Schackmuth MK, Shiers S, Varela D, Price TJ, Schmidt M (2025). Multi-omic integration with human DRG proteomics highlights TNF α signalling as a relevant sexually dimorphic pathway. *PAIN* ():10.1097/j.pain.0000000000003656, May 20, 2025. **Editor's choice.**

Xian F, Brenek M, Krisp C, Aguanno D, Urbauer E, Srikumar T, Ravi Kumar RK, Liu Q, **Barry AM**, Ma B, Krieger J, Haller D, Schmidt M, Gomez Varela D (2025). Ultra-sensitivity metaproteomics redefines the gut “dark metaproteome”, uncovering host-microbiome interactions and drug targets in intestinal inflammatory diseases. *Nat Commun* **16**, 6644. <https://doi.org/10.1038/s41467-025-61977-7>

Zhao N, Bennett DL, Baskozos G, **Barry AM** (2024). Predicting “pain genes”: multi-modal data integration using probabilistic classifiers and interaction networks. *Bioinformatics Advances*, <https://doi.org/10.1093/bioadv/vbae156>

Lacagnina MJ, Willcox KF, Boukelmoune N, Bavencoffe A, Sankaranarayanan I, Barratt DT, Zuberi YA, Dayani D, Chavez MV, Lu JT, Farinotti AB, Shiers S, **Barry AM**, Mwirigi JM, Tavares-Ferreira D, Funk GA, Cervantes AM, Svensson CI, Walters ET, Hutchinson MR, Heijnen CJ, Price TJ, Fiore NT, Grace PM (2024). B cells drive neuropathic pain-related behaviors in mice through IgG–Fc gamma receptor signaling. *Sci Transl Med*. 2024. 16(766) DOI: 10.1126/scitranslmed.adj1277

Farah A, Patel R, Poplawski P, Wastie BJ, Tseng M, **Barry AM**, Daifallah O, Dubb A, Paul I, Cheng HL, Feroz F, Su Y, Chan M, Zeilhofer HU, Price T, Bennett DL, Bannister K, Dawes JM (2024). A role for leucine-rich, glioma inactivated 1 in regulating pain sensitivity. *Brain*. 2024. Sep 20;awae302. doi: 10.1093/brain/awae302. PMID: 39301592.

Cooper AH, **Barry AM**, Chrysostomidou P, Lolignier R, Titterton HF, Bennett DL, Weir GA (2024). Peripheral nerve injury results in a biased loss of sensory neuron sub-populations. *PAIN*. 10.1097/j.pain.0000000000003321

Barry AM, Zhao N, Yang X, Bennett DL, Baskozos G (2022). Deep RNA-seq of male and female murine sensory neuron subtypes after nerve injury. *PAIN*. 2023. 164(10):2196-2215. **Editor's choice.**

Middleton SJ, Perini I, Themistocleous AC, Weir GA, McCann K, **Barry AM**, Marshall A, Lee M, Mayo LM, Bohic M, Baskozos G, Morrison I, Löken LS, McIntyre S, Nagi SS, Staud R, Sehlstedt I, Johnson RD, Wessberg J, Wood JN, Woods CG, Moqrich A, Olausson H, Bennett DL (2022). Nav1.7 is required for normal C-low threshold mechanoreceptor function in humans and mice. *Brain*. Oct 21;145(10):3637-3653. doi: 10.1093/brain/awab482. PMID: 34957475; PMCID: PMC9586547.

Middleton SJ*, **Barry AM***, Comini M, Li Y, Ray PR, Shiers S, Themistocleous AC, Uhelski ML, Yang X, Dougherty PM, Price TJ, Bennett DL (2021). Studying human nociceptors: from fundamentals to clinic. *Brain*. 144(5):1312–1335. <https://doi.org/10.1093/brain/awab048> *shared first authorship

Sondermann JR, **Barry AM**, Jahn O, Michel N, Abdelaziz R, Kügler S, Gomez-Varela D, Schmidt M. Vt1b promotes TRPV1 sensitization during inflammatory pain. *PAIN*. 2019. Feb;160(2):508-527. doi: 10.1097/j.pain.0000000000001418. PMID: 30335684.

Barry AM, Sondermann JR, Sondermann J-H, Gomez-Varela D, Schmidt M (2018) Region-resolved quantitative proteome profiling reveals molecular dynamics associated with chronic pain in the PNS and spinal cord. *Frontiers in Molecular Neuroscience* 11 (August): 259.

Gomez-Varela D, **Barry AM**, Schmidt M (2019). Proteome-based systems biology in chronic pain. *J Proteomics*. Jan 6;190:1-11. doi: 10.1016/j.jprot.2018.04.004.

Other Publications (eg. protocols, datasets)

Barry A, Schmidt M, Price TJ (2026). Bulk proteomics of human dorsal root ganglia from subjects with mixed medical histories (Version 1) [Dataset]. SPARC Portal. <https://doi.org/10.26275/HXNX-TSEC>

Barry A, Schmidt M, Price TJ (2025). Bulk proteomics of naive human dorsal root ganglion (Version 1) [Dataset]. SPARC Portal. <https://doi.org/10.26275/Z7UY-KUIF>

Barry A, Price TJ, Schmidt M (2024). Bulk Proteomics (DIA-MS) of Human Dorsal Root Ganglion v1. <https://doi.org/10.17504/protocols.io.j8nlk8j56l5r/v1>

Barry AM (2022). Exploring the molecular underpinnings of neuropathic pain in primary afferent subtypes. (Doctoral dissertation, University of Oxford)